

Xiaoling Hu

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Mobile: 6312028413, *Google Scholar* (1400+ citations): <https://tinyurl.com/39yvajec/>

Current Position • **Harvard University, USA** Aug. 2023 - Present
 Postdoctoral Research Fellow

Research Interests My research interests lie in machine learning, computer vision, with applications in health-care, multimodal AI, generative AI, VLM/LLMs, and trustworthy learning systems. In particular, I am interested in:

- **Topological Deep Learning**
- **Trustworthy Machine Learning**
- **Multimodal AI and Generative AI (GenAI)**
- **VLM/LLM Reasoning, Efficiency and Confidence**

Education • **Department of CS, Stony Brook University, USA** Jan. 2018 - June 2023
 Doctor of Philosophy

 • **Department of EE, Tsinghua University, China** Sep. 2014 - June 2017
 Master of Engineering

 • **Department of EE, Huazhong University of Science and Technology, China**
 Sep. 2010 - June 2014
 Bachelor of Science

Selected Honors and Awards

- CVPR Oral (Top 1%) 2025
- Catecosinos Fellowship 2023
- ICLR Spotlight (Top 5%) 2023
- ECCV Oral (Top 2%) 2022
- ICLR Spotlight (Top 5%) 2021
- NeurIPS Travel Award 2019

Professional Service **Area Chair**

- Conference on Neural Information Processing Systems (NeurIPS) 2025, 2026
- International Conference on Learning Representations (ICLR) 2027
- The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026, 2027
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2026, 2027
- AAAI Conference on Artificial Intelligence 2027
- The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2027
- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2025, 2026

Selected Publications (* indicates equal contribution, — denotes students (co-)mentored by me, ‡ denotes (co)-senior supervision)

- [1] **Multi-Channel Uncertainty-Weighted Score Matching for Conditional Diffusion in Medical UDA**
Chen Li, Meilong Xu, **Xiaoling Hu**, Weimin Lyu, Chao Chen
European Conference on Computer Vision (ECCV), 2026
- [2] **Act Like a Pathologist: Tissue-Aware Whole Slide Image Reasoning**
Wentao Huang, Weimin Lyu, Peiliang Lou, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Wenchao Han, Ruifeng Guo, Jiawei Zhou, Chao Chen, Chen Wang
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- [3] **MATCH: Multi-faceted Adaptive Topo-Consistency for Semi-Supervised Histopathology Segmentation**
Meilong Xu*, **Xiaoling Hu***, Shahira Abousamra, Chen Li, Chao Chen
Thirty-Ninth Conference on Neural Information Processing Systems (NeurIPS), 2025
Short version is selected as **Oral Presentation** at Imageomics Workshop at NeurIPS 2025
- [4] **Learn2Synth: Learning Optimal Data Synthesis Using Hypergradients for Brain Image Segmentation**
Xiaoling Hu, Xiangrui Zeng, Oula Puonti, Juan Eugenio Iglesias, Bruce Fischl‡, Yaël Balbastre‡
International Conference on Computer Vision (ICCV), 2025
- [5] **TopoCellGen: Generating Histopathology Cell Topology with a Diffusion Model**
Meilong Xu, Saumya Gupta, **Xiaoling Hu**, Chen Li, Shahira Abousamra, Dimitris Samaras, Prateek Prasanna, Chao Chen
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025 (**Oral Presentation, acceptance rate < 1%**)
Short version is also presented at CVPR 2025 Workshop GMCV
- [6] **Hierarchical Uncertainty Estimation for Learning-based Registration in Neuroimaging**
Xiaoling Hu, Karthik Gopinath, Peirong Liu, Malte Hoffmann, Koen Van Leemput, Oula Puonti‡, Juan Eugenio Iglesias‡
International Conference on Learning Representations (ICLR), 2025
- [7] **Semi-supervised Segmentation of Histopathology Images with Noise-Aware Topological Consistency**
Meilong Xu, **Xiaoling Hu**, Saumya Gupta, Shahira Abousamra, Chao Chen
European Conference on Computer Vision (ECCV), 2024
- [8] **Brain-ID: Learning Contrast-agnostic Anatomical Representations for Brain Imaging**
Peirong Liu, Oula Puonti, **Xiaoling Hu**, Daniel C. Alexander, Juan Eugenio Iglesias
European Conference on Computer Vision (ECCV), 2024
- [9] **Topology-Aware Uncertainty for Image Segmentation**
Saumya Gupta, Yikai Zhang, **Xiaoling Hu**, Prateek Prasanna, Chao Chen
Thirty-Seventh Conference on Neural Information Processing Systems (NeurIPS), 2023

- [10] **Calibrating Uncertainty for Semi-Supervised Crowd Counting**
Chen Li, **Xiaoling Hu**, Shahira Abousamra, Chao Chen
International Conference on Computer Vision (ICCV), 2023
- [11] **Enhancing Modality-Agnostic Representations via Meta-Learning for Brain Tumor Segmentation**
Aishik Konwer, **Xiaoling Hu**, Xuan Xu, Joseph Bae, Chao Chen, Prateek Prasanna
International Conference on Computer Vision (ICCV), 2023
- [12] **Learning Probabilistic Topological Representations Using Discrete Morse Theory**
Xiaoling Hu, Dimitris Samaras, Chao Chen
International Conference on Learning Representations (ICLR), 2023 (**Spotlight Presentation, notable-top-25% of accepted papers**)
*Short version is selected as **Oral Presentation** at Medical Imaging meets NeurIPS Workshop, 2023*
- [13] **Confidence Estimation Using Unlabeled Data**
Chen Li, **Xiaoling Hu**, Chao Chen
International Conference on Learning Representations (ICLR), 2023
- [14] **Structure-Aware Image Segmentation with Homotopy Warping**
Xiaoling Hu
Thirty-Sixth Conference on Neural Information Processing Systems (NeurIPS), 2022
- [15] **Learning Topological Interactions for Multi-Class Medical Image Segmentation**
 Saumya Gupta*, **Xiaoling Hu***, James Kaan, Michael Jin, Mutshipay Mpoy, Katherine Chung, Gagandeep Singh, Mary Saltz, Tahsin Kurc, Joel Saltz, Apostolos Tassiopoulos, Prateek Prasanna, Chao Chen
European Conference on Computer Vision (ECCV), 2022 (**Oral Presentation, acceptance rate 2.7%**)
- [16] **Trigger Hunting with a Topological Prior for Trojan Detection**
Xiaoling Hu, Xiao Lin, Michael Cogswell, Yi Yao, Susmit Jha, Chao Chen
International Conference on Learning Representations (ICLR), 2022
- [17] **Topology-Aware Segmentation Using Discrete Morse Theory**
Xiaoling Hu, Yusu Wang, Li Fuxin, Dimitris Samaras, Chao Chen
International Conference on Learning Representations (ICLR), 2021 (**Spotlight Presentation, acceptance rate 5.6%**)
- [18] **Topology-Preserving Deep Image Segmentation**
Xiaoling Hu, Li Fuxin, Dimitris Samaras, Chao Chen
Thirty-Third Conference on Neural Information Processing Systems (NeurIPS), 2019

Preprints

- [1] **Learning to Upscale 3D Segmentations in Neuroimaging**
Xiaoling Hu, Peirong Liu, Dina Zemlyanker, Jonathan Williams Ramirez, Oula Puonti, Juan Eugenio Iglesias
Tech Report, 2025
- [2] **ORCE: Order-Aware Alignment of Verbalized Confidence in Large Language Models**
Chen Li, **Xiaoling Hu**, Songzhu Zheng, Jiawei Zhou, Chao Chen
Tech Report, 2026

- [3] **Topo-R1: Detecting Topological Anomalies via Vision-Language Models**
 Meilong Xu, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Xin Yu, Weimin Lyu,
 Kehan Qi, Dimitris Samaras, Chao Chen
Tech Report, 2026
- [4] **Unrolled Networks are Conditional Probability Flows in MRI Reconstruction**
 Kehan Qi, Saumya Gupta, **Xiaoling Hu**, Qingqiao Hu, Weimin Lyu, Chao Chen
Tech Report, 2025
- [5] **LoC-Path: Learning to Compress for Pathology Multimodal Large Language Models**
 Qingqiao Hu, Weimin Lyu, Meilong Xu, Kehan Qi, **Xiaoling Hu**, Saumya Gupta,
 Jiawei Zhou, Chao Chen
Tech Report, 2025
- [6] **Uncertainty Estimation for Pretrained Medical Image Registration Models via Transformation Equivariance**
 Lin Tian, **Xiaoling Hu**, Juan Eugenio Iglesias
Tech Report, 2025

Mentoring

- Chen Li (**ICLR'23, ICCV'23, MICCAI'24, ECCV'26**), Ph.D. student at Department of BMI, Stony Brook University
 Since Fall 2021
- Meilong Xu (**ECCV'24, CVPR'25 Oral Presentation, NeurIPS'25**), Ph.D. student at Department of CS, Stony Brook University
 Since Summer 2023
- Wentao Huang (**MICCAI'24, CVPR'26, TMI'26**), Ph.D. student at Department of CS, Stony Brook University
 Since Summer 2023
- Qingqiao Hu, Ph.D. student at Department of CS, Stony Brook University
 Since Fall 2024
- Kehan Qi, Ph.D. student at Department of BMI, Stony Brook University
 Since Fall 2025
- Jiaqi Yang (**MICCAI'21, ISBI'21, MedIA'24, JBHI'25**), Ph.D. student at Department of CS, CUNY
 Spring 2020 – Summer 2023
- Saumya Gupta (**ECCV'22 Oral Presentation, NeurIPS'23**), Ph.D. student at Department of CS, Stony Brook University
 Fall 2021 – Summer 2023
- Mahmudul Hasan (**MICCAI'24**), Ph.D. student at Department of CS, Stony Brook University
 Summer 2023 – Summer 2024
- Luca Drole, Master student in BME, ETH Zurich
 Spring - Fall 2025
- John Xie, High School student → University of Michigan
 Summer 2021

Talks

Principled Learning for Medical AI: Structure, Reliability, and Interpretability

- School of Computing & SCI Institute, University of Utah Mar. 2026
- School of Computing, University of Georgia Mar. 2026
- Department of EECS, University of Tennessee, Knoxville Feb. 2026
- Department of CS, Missouri S&T Jan. 2026
- HIT Webinar - *Healthcare, Intelligence, Technology* Aug. 2025
- Department of CSE, University of North Texas Nov. 2025

***Learn2Synth*: A Learnable Data Synthesis Strategy for Image Segmentation**

- Nobrainer Seminar, Massachusetts Institute of Technology June 2024

Deep Structural Reasoning for Biomedical Imaging

- School of CAI, Arizona State University Feb. 2024

Topology-Aware Deep Image Segmentation

- MICCAI'23 tutorial on *Topology-Driven Image Analysis*, Vancouver Oct. 2023

Learning Topological Representations for Deep Image Understanding

- Department of CS, Florida State University Apr. 2023
- Department of BMI, Ohio State University Mar. 2023
- Department of CS, Rochester Institute of Technology Feb. 2023
- Department of ECE, University of California, Riverside Feb. 2023
- Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital & Harvard Medical School Nov. 2022

Learning Probabilistic Topological Representations Using Discrete Morse Theory

- Medical Imaging meets NeurIPS Workshop, New Orleans Dec. 2022

Topology-Informed Image Analysis

- Center for Computational Neuroscience, Flatiron Institute Oct. 2022

Topology-Aware Deep Image Segmentation

- Geometry and Topology meet Data Analysis and Machine Learning Aug. 2021

Topology-aware Segmentation Using Discrete Morse Theory

- International Conference on Learning Representations (ICLR) May 2021

**Industry
Experiences**

- **Allen Institute, USA**

Research Intern

Mentor: *Dr. Matheus Viana*

Topic: Topology-Aware Image Segmentation

May 2022 - Aug. 2022

- **United Imaging Intelligence (UII), USA**

Research Intern

Mentor: *Dr. Shanhui Sun*

Topic: Deep Shape Model Based Network

May 2021 - Aug. 2021

- **Tencent Youtu Lab, China**

Research Intern

Mentor: *Dr. Yuwing Tai*

Topic: Clothes Detection, Attribute Prediction

Jun. 2017 - Jan. 2018